

RUCHIR SHUKLA

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Education

B.Tech in Electronics and Telecommunication Engineering

Sardar Patel Institute of Technology

Oct 2022 - Jul 2026

CGPA: 9.22

Minor in Computer Engineering

Sardar Patel Institute of Technology

Aug 2023 - Dec 2025

Work Experience

Application Developer Intern - Russell Investments (Ops - Tech)

May 2025 - Aug 2025

- Designed & developed modules automating backend data analysis jobs to reduce manual efforts, improve efficiency & scalability
- Created in-house tools using **Python** and **.NET** to replace commercial software, **improving performance and enabling greater customization.**
- Revamped existing API's by eliminating dependencies on paid legacy software to increase maintainability
- Performed backend API testing using **Swagger** for functional validation and **Locust** for simulating concurrent load and stress conditions.

Projects

Samarth AI 🌀 (Smart India Hackathon 24) | *Python, FastAPI, ChromaDB, React*

Dec 2024

- Developed a ChatBot to aid employees search and understand technical documents and HR policies using NLP, thereby boosting their productivity.
- Implemented **RAG model** to retrieve relevant information from lengthy documentation and policies and generate response using Llama3 8B.
- Split the information into chunks and added **Contextual Chunk Headers** to improve accuracy.
- Used **Key Cycling** and **Multi-Threading** to reduce initial processing time of documents.
- Collaborated with team to ensure seamless integration of all features.

Outfit Recommender 🌀 | *TensorFlow, Pandas, Django, React*

May 2024

- Developed a clothing e-commerce platform with a personalized outfit recommendation system.
- Used **Pandas** to clean the dataset and reduce size by uniform sampling.
- Implemented **Collaborative Filtering** using **TensorFlow** to generate clothing recommendations.
- Aimed to enhance user experience, boost cross-sell and customer retention through real-time recommendations.

OptimoRoute 🌀 | *Python, Django, React, MongoDB*

Apr 2024

- Developed a web application to compare trade routes across land, air, and sea using data from **SeaRoutes** and **GraphHopper APIs**.
- Retrieved and analyzed distance, estimated time, and CO2 emissions for each transportation mode to assist in eco-conscious decision-making.
- Handled backend logic and **REST API** integration with **Django**.

OctoChip 🌀 | *Logisim, Microprocessors*

May 2024

- Designed & developed a 8-bit RISC processor ("OctoChip") using LogiSim Application (Logic circuit simulator).
- Implemented 32 instructions (expandable to 64) enabling the user to load programs for complex computations.
- OctoChip computes with the help of 256B RAM, 4 General Purpose registers, softwired instruction decoder.

Technical Skills

Languages: Python, Java, C++, .Net, Assembly, MicroPython

Web Development: Django, FastAPI, REST API

Development Tools: Github, Postman, Swagger, Locust

Achievements

Smart India Hackathon 2024 - *First place among 150+ teams from all over India*

Dec 2024

C.S.I. Takedown - *Secured 2nd place in college level Coding Event competing in pairs against 150 teams*

Jun 2023

C.S.I. Pixel Perfect - *Secured 3rd Place in college level UI Design Competition*

May 2023

Positions Of Responsibility

Finance Lead - Students' Developers Club

Aug 2024 - May 2025

- Led the planning and execution of technical events and hackathons as part of the SDC Super-core team.

Training & Placement Coordinator - S.P.I.T, Placement Office

Aug 2023 - Oct 2024

- Placement drives - Collaborated with the companies & assisted students in campus recruitment drives